

Solana Stablecoin Standard Bounty Submission Audit Report

Generated: February 25, 2026

Network: Solana Devnet | 28/28 Tests Passing

Component	Program ID
Stablecoin Program	GPXDvDTpDnCxWrkKXYkfFedKWhsvbmLj2FpXNQM3EV7y
Transfer Hook Program	C76nk4L27JJbXiVHR72mWdcq9jX8NETHeKECAxw72ZpM
Authority	Ek73zmhXDLiKi6dca91RDUrmj8TdobZYunmUrCJecuti

1. On-Chain Program (Anchor)

Single configurable Anchor program supporting SSS-1 (Minimal) and SSS-2 (Compliant) presets via initialization parameters. Separate transfer hook program for blacklist enforcement.

Core Instructions (All Presets)

Instruction	Status	Source File
initialize	PASS	instructions/initialize.rs (256 lines)
mint_tokens	PASS	instructions/mint.rs
burn_tokens	PASS	instructions/burn.rs
freeze_account	PASS	instructions/freeze.rs
thaw_account	PASS	instructions/thaw.rs
pause / unpause	PASS	instructions/pause.rs, unpause.rs
manage_role (grant/revoke)	PASS	instructions/roles.rs
transfer_authority	PASS	lib.rs + TransferAuthority ctx

SSS-2 Compliance Instructions

Instruction	Status	Source File
add_to_blacklist	PASS	instructions/blacklist.rs
remove_from_blacklist	PASS	instructions/blacklist.rs (PDA close)
seize (burn+mint)	PASS	instructions/seize.rs (139 lines)
transfer hook (blacklist)	PASS	programs/transfer-hook/src/lib.rs

Role-Based Access Control

Role	Preset	Description
Minter	SSS-1+	Mint tokens, per-minter quotas
Burner	SSS-1+	Burn tokens
Pauser	SSS-1+	Pause/unpause all operations
Blacklister	SSS-2	Add/remove blacklist entries
Seizer	SSS-2	Seize tokens from blacklisted accounts

Feature gating: SSS-2 instructions (blacklist, seize) return `ComplianceNotEnabled` error when called on SSS-1 mints. Verified in devnet tests 25-26.

2. Devnet Deployment Proof

Fresh deployment with new keypairs. All 28 tests pass with on-chain transaction signatures.

SSS-1: Minimal Stablecoin (12/12)

Test	Status	Tx Signature (prefix)
Initialize SSS-1 stablecoin	PASS	5GYVcrFhom3Ecz...
Config matches preset	PASS	(verified on-chain)
Mint 10 tokens to ATA	PASS	afUE3rbo4CPJ4x...
Burn 1 token (9 remaining)	PASS	2c6fxjzGf1osPT...
Freeze account	PASS	2uCeWVGGPZfV9j...
Thaw account	PASS	hhQpSbqdYY3KTZ...
Pause stablecoin	PASS	4GNtunHDNffzWh...
Mint rejected while paused	PASS	(error expected)
Unpause stablecoin	PASS	5zCA7ZZhdC6cu1...
Grant Minter role	PASS	VgexbiqbYP4pc9...
Revoke Minter role	PASS	5BVAAEUan6vVB3...
Token-2022 transfer (no hook)	PASS	4hNPFudXxK68Vu...

SSS-2: Compliant Stablecoin (10/10)

Test	Status	Tx Signature (prefix)
Initialize SSS-2 stablecoin	PASS	3JBsBgHkHmK8Lu...
Init transfer hook extra-account-metas	PASS	5Bpn26jYGk5wVK...
Config: freeze=true, delegate=true, hook=true	PASS	(verified on-chain)
Mint 50 SSS-2 tokens	PASS	3hwAGbjZTqXxYG...
Add address to blacklist	PASS	35U7pv4H6yB2NE...
Duplicate blacklist rejected	PASS	(error expected)
Seize tokens (5 tokens)	PASS	48AKmDvJ9RmxG8...
Seize rejected without Seizer role	PASS	(error expected)
Remove from blacklist (PDA closed)	PASS	3EwhTwVTLJKtho...
Blacklist rejected on SSS-1 mint	PASS	(error expected)

Role Separation + Authority Transfer (6/6)

Test	Status
Grant Blacklister role	PASS
Grant Seizer role	PASS
Seizer role rejected on SSS-1	PASS
Blacklister role rejected on SSS-1	PASS
Transfer authority to new keypair	PASS
Old authority rejected after transfer	PASS

3. TypeScript SDK

SDK Components

Component	File	Description
SolanaStablecoin class	sdk/src/stablecoin.ts (239 lines)	create(), load(), mint(), burn(), pause(), unpause()
Token operations	sdk/src/instructions/token-ops.ts	mint, burn instruction builders
Compliance module	sdk/src/instructions/compliance.ts	blacklistAdd, blacklistRemove, seize
Role manager	sdk/src/instructions/roles.ts	grant, revoke role instructions
Types + presets	sdk/src/types.ts (139 lines)	Presets enum, FeatureFlags, StablecoinConfig
PDA helpers	sdk/src/utlis/pda.ts	findConfigPDA, findRolePDA, findBlacklistPDA
IDL (patched)	sdk/src/idl/solana_stablecoin.json	12 instructions, 3 accounts, discriminators

4. Admin CLI

CLI Commands

Command	Status	Source
sss-token init --preset sss-1/sss-2	IMPL	cli/src/commands/init.ts
sss-token mint / burn	IMPL	cli/src/commands/mint.ts, burn.ts
sss-token freeze / thaw	IMPL	cli/src/commands/freeze.ts
sss-token pause / unpause	IMPL	cli/src/commands/pause.ts
sss-token status / supply	IMPL	cli/src/commands/status.ts
sss-token blacklist add/remove	IMPL	cli/src/commands/blacklist.ts
sss-token seize	IMPL	cli/src/commands/blacklist.ts
sss-token roles grant/revoke/list	IMPL	cli/src/commands/roles.ts
sss-token holders	IMPL	cli/src/index.ts -> statusCommand
sss-token audit-log	IMPL	cli/src/index.ts -> statusCommand

5. Backend Services (Docker)

Containerized Services

Service	Port	Dockerfile	Health Check
Mint/Burn Service	3001	services/mint-burn/Dockerfile	/health
Event Indexer	3002	services/indexer/Dockerfile	/health
Compliance Service (SSS-2)	3003	services/compliance/Dockerfile	/health
Webhook Service	3004	services/webhook/Dockerfile	/health

All services configured in `docker-compose.yml` with `env_file`, `healthchecks` (30s interval), and `restart` policies.

6. Documentation

Required Documents

Document	Status	Lines	Contents
README.md	DONE	185	Overview, quick start, preset comparison
docs/ARCHITECTURE.md	DONE	165	Layer model, data flows, security
docs/SDK.md	DONE	120	Presets, custom configs, TS examples
docs/OPERATIONS.md	DONE	117	Operator runbook
docs/SSS-1.md	DONE	67	Minimal stablecoin spec
docs/SSS-2.md	DONE	91	Compliant stablecoin spec
docs/COMPLIANCE.md	DONE	90	Regulatory considerations, audit trail
docs/API.md	DONE	173	Backend API reference

7. Tests

Test Coverage

Test Type	Status	File / Evidence
Integration tests (33 cases)	DONE	tests/integration/stablecoin.test.ts (1021 lines)
Devnet verification (28 cases)	PASS	testnet-test.ts — all tx signatures on-chain
Fuzz tests (Trident)	DONE	tests/fuzz/fuzz_compliance.rs, fuzz_roles.rs
Preset config tests	DONE	6 preset tests in integration suite
SSS-1 flow: mint->transfer->freeze	PASS	Devnet tests 1-12
SSS-2 flow: mint->blacklist->seize	PASS	Devnet tests 13-22

Summary

Category	Required	Delivered	Status
On-chain program (instructions)	12	12	COMPLETE
Transfer hook program	1	1	COMPLETE
Devnet tests passing	All	28/28	COMPLETE

TypeScript SDK	1	1 (7 modules)	COMPLETE
Admin CLI	1	1 (10+ commands)	COMPLETE
Backend services	4	4 (Dockerized)	COMPLETE
Documentation	8 docs	8 docs	COMPLETE
Tests (integration + fuzz)	Required	33 + 2 fuzz	COMPLETE
Docker setup	Required	docker-compose.yml	COMPLETE

All submission requirements met. Source code, working tests, devnet deployment proof (program IDs + example transactions), documentation, and Docker setup included.